

Drake Ferronyalka

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PROFESSIONAL SUMMARY

Mechanical Engineering student and U.S. Navy veteran who takes designs from concept through fabrication and validation, across R&D prototyping, precision machining, and mechanical design. Proficient in SolidWorks (CSWA Certified), GD&T, and DFM/DFA, with a disciplined approach to requirements and documentation. Available onsite, full-time for 12 consecutive weeks beginning May 2027; open to any site.

EDUCATION

University of Central Florida

Jan 2026 – Present

B.S. Mechanical Engineering (Expected Graduation: May 2028; GPA: 3.85)

State College of Florida

Dec 2025

A.A. President's List (Fall 2024, Spring/Fall 2025); Dean's List (Spring/Summer 2024)

EXPERIENCE

Machinist Apprentice

May 2026 – Present

University of Central Florida, Office of Research

Orlando, FL

- Fabricate components in direct support of Senior Design teams and faculty-led research labs, applying first-principles reasoning to translate design intent into producible geometry under real shop constraints.
- Produce GD&T drawings and SolidWorks designs for shop fixtures and CNC tooling used across multiple concurrent research and student projects.
- Advise students and research teams on design-for-manufacturability, reducing rework and improving part producibility on available shop equipment.

Mechanical Design and Fabrication Technician

2020 – Present

AFNY Specialties (2020 – 2025) / Pulsar Designs LLC (2025 – Present)

Ellenton, FL

- Worked remotely while on active duty through 2023, then on-site part-time while completing an A.A.; designed and fabricated SolidWorks assemblies ranging from single-piece brackets to 10-15 part mounts and suspension hardware engineered for high-vibration duty; produced BOMs and assembly notes supporting fabrication and installation.
- Standardized repeatable threaded features using a FlexArm tapping system, optimizing tap geometry, torque control, and lubrication practices to reduce tool breakage and maintain thread quality.

Fire Controlman / 400 Hz Power System Maintainer

2017 – 2023

United States Navy

Various Locations

- Maintained and troubleshot MK 99 T-1348 Radar Transmitter and ACSSFC 400 Hz power systems aboard DDG-class ships, executing preventive and corrective maintenance to sustain operational readiness in high-tempo environments.
- Led and trained 10+ technicians on diagnostics, troubleshooting workflows, and procedure compliance; standardized checkouts and technical direction improved team execution.

PROJECTS

X-Axis Linear Rail Conversion – CNC Plasma Table

Jan 2026 – May 2026

SolidWorks, Hand Calculations, Linear Rail Systems, Machining

- Converted the X-axis of a Langmuir CrossFire XR plasma table from a hand-preloaded ball-bearing carriage to an HGR20 profile rail, replacing a by-feel preload adjustment that became sticky and sporadic at low shop temperatures.
- Sized a two-part 6061-T6 bracket by hand calculation from measured loads (25.31 N Z-axis assembly, 22.27 N cableway), reacting a 1.974 N·m root moment as a force couple across two HGH20CA blocks at 78.7 mm centers, the minimum achievable spacing for the block size.
- Validated geometry, load path, and rail mounting with an FDM PETG printed article which ran in production for two months before committing to billet; machined components were installed May 2026 and in continuous production service with no preload corrections.

Wheel Light Mounting System (US 12,673,603 B2 – granted Jul 2026)

2024 – 2025

SolidWorks, GD&T, Prototype Fabrication

- Designed a modular mounting architecture for off-road lighting with emphasis on vibration resistance, packaging, and serviceability as a named co-inventor; produced manufacturing-ready drawings and BOM.
- Fabricated prototypes via laser-cut flat patterns, formed and assembled in-house; validated performance via torque checks and road testing.

SKILLS

CAD & Design: SolidWorks (CSWA Certified), Siemens NX (modeling), GD&T (ASME Y14.5), Technical Drawing, Tolerance Stack-Up, DFM/DFA

Analysis: ANSYS Static Structural, First-Principles Sizing, Shaft & Gear Load Analysis

Manufacturing: Manual Lathe & Mill, CNC 3-Axis Mill (setup & operation), CNC Plasma (SheetCAM), Sheet Metal/Plate Fabrication, Press Brake Forming, Tapping/Threaded Features, Fixture Design, FDM, Assembly & Integration

Instrumentation & Electronics: BLDC Motor Control (VESC), Data Acquisition, Load Cells, Thermocouples, IMU, Oscilloscope, Soldering, Radar & 400 Hz Power Systems

Software: Python, C, Embedded (RP2040 / Raspberry Pi), Microsoft Excel/Word/PowerPoint